

Fraud Detection in Financial Transactions Using ML

An Analysis of the PaySim Synthetic Financial Dataset

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Abstract

Financial fraud poses a significant threat to the integrity of digital payment systems, resulting in substantial economic losses worldwide. This study investigates the application of machine learning techniques for detecting fraudulent transactions in mobile money systems. Using the PaySim synthetic financial dataset, which simulates real-world mobile money transactions, we aim to develop and evaluate classification models capable of accurately identifying fraudulent activities. The dataset contains over 6.3 million transactions across five transaction types (CASH_IN, CASH_OUT, DEBIT, PAYMENT, and TRANSFER), with approximately 8,213 labeled fraudulent cases representing a highly imbalanced classification problem. Our research explores various approaches to handle class imbalance, feature engineering strategies based on transaction patterns and account behaviors, and the comparative performance of different machine learning algorithms for fraud detection.

Research Questions

1. Which machine learning algorithms perform best for detecting fraud in imbalanced financial transaction data?
2. What transaction features are most predictive of fraudulent behavior?
3. How can class imbalance be effectively addressed to minimize false negatives while maintaining acceptable precision?

Dataset Description

Name: PaySim Synthetic Financial Dataset

Source: Kaggle - <https://www.kaggle.com/datasets/ealaxi/paysim1>

Description:

PaySim is a financial simulator that generates synthetic data based on a sample of real transactions extracted from one month of financial logs from a mobile money service in Africa. The dataset simulates mobile money transactions to facilitate fraud detection research.

Dataset Statistics:

Attribute	Details
Total Records	6,362,620 transactions
Features	11 columns
Fraudulent Cases	8,213 (0.13%)
Transaction Types	CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER

Features:

- **step:** Unit of time (1 step = 1 hour)
- **type:** Transaction type (CASH_IN, CASH_OUT, DEBIT, PAYMENT, TRANSFER)
- **amount:** Transaction amount
- **nameOrig:** Customer initiating the transaction
- **oldbalanceOrg / newbalanceOrig:** Balance before/after transaction (origin)
- **nameDest:** Recipient of the transaction
- **oldbalanceDest / newbalanceDest:** Balance before/after transaction (destination)
- **isFraud:** Target variable (1 = fraud, 0 = legitimate)
- **isFlaggedFraud:** System flagged as fraud attempt